

Drivers of agile software development use: Dialectic interplay between benefits and hindrances

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ABSTRACT

Context: Agile software development with its emphasis on producing working code through frequent releases, extensive client interactions and iterative development has emerged as an alternative to traditional plan-based software development methods. While a number of case studies have provided insights into the use and consequences of agile, few empirical studies have examined the factors that drive the adoption and use of agile.

Objective: We draw on intention-based theories and a dialectic perspective to identify factors driving the use of agile practices among adopters of this software development methodology.

Method: Data for the study was gathered through an anonymous online survey of software development professionals. We requested participation from members of a selected list of online discussion groups, and received 98 responses.

Results: Our analyses reveal that subjective norm and training play a significant role in influencing software developers' use of agile processes and methods, while perceived benefits and perceived limitations are not primary drivers of agile use among adopters. Interestingly, perceived benefit emerges as a significant predictor of agile use only if adopters face hindrances to their agile practices.

Conclusion: We conclude that research in the adoption of software development innovations should examine the effects of both enabling and detracting factors and the interactions between them. Since training, subjective norm, and the interplay between perceived benefits and perceived hindrances appear to be key factors influencing the adoption of agile methods, researchers can focus on how to (a) perform training on agile methods more effectively, (b) facilitate the dialog between developers and managers about perceived benefits and hindrances, and (c) capitalize on subjective norm to publicize the benefits of agile methods within an organization. Further, when managing the transition to new software development methods, we recommend that practitioners adapt their strategies and tactics contingent on the extent of perceived hindrances to the change.

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1. Introduction

Agile development tends to focus on early and fast production of working code, frequent, small, incremental changes, pair programming, short iterations, rapid and continual user feedback and interaction [1,2]. In contrast, traditional development tends to emphasize extensive analysis before coding, creation and maintenance of models, relatively long time spans between deliverables and milestones, and typically small amounts of user interaction [3,4]. While there is anecdotal evidence supporting the successes of both approaches, it appears that organizations are increasingly concluding that, because of the recurring problem of uncertain or unstable requirements and the production of inflexible systems

that frequently fail to meet user needs, it is best to avoid strictly phased or non-iterative processes [5].

However, while most organizations may accept that strictly linear, phased, and non-iterative approaches to software development, such as the systems development life cycle (SDLC) or traditional waterfall model, are usually not the most effective to follow, there are few empirical studies that have researched the adoption, use and consequences of different software development methods utilizing larger data samples that could provide guidance to practitioners who have to decide which techniques and approaches to use. Moreover, this general lack of empirical data makes it difficult to reliably forecast the expected bottom-line results (cost, time, quality, validity) that may be expected if a given software development approach is adopted. In fact, Dybå and Dingsøyr [6], in a critical review of existing empirical studies on agile development, concluded that more rigorous studies needed to be conducted before any credible advice can be given to software

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development practitioners regarding the use of agile approaches. Such studies could offer more generalizable results that corroborate or refute the anecdotal evidence, and, more importantly, offer practical guidelines that can assist software developers, managers and executives make more informed decisions in their selection of software development approaches.

The objective of this study is to examine the perceptions of adopters of agile development techniques in order to better understand the factors influencing their usage decisions. We grounded our study on extant literature by drawing on the theories of reasoned action and planned behavior that have traditionally guided research on the adoption and use of software development innovations. Accordingly, we modeled and tested the use of agile methodologies as a function of salient and normative beliefs about its perceived benefits and limitations.

Interestingly, as will be discussed below, we found that a dialectic perspective regarding these perceived benefits and limitations may help explain adoption and give managers guidance regarding how to select a software development approach and in how to facilitate the adoption of the selected approach. The dialectic perspective suggests that it is the interplay between perceived benefits and limitations – not just the existence of perceived benefits – that determines their importance in the acceptance and use of agile methodologies. It is through rational discussion about the perceived benefits and limitations that disagreement may be resolved and that an appropriate development methodology may emerge and be adopted.

Our study, with data collected from a relatively large group of respondents, aims to contribute to theory and practice by shedding light on the key enabling and detracting factors and the interactions between them in the context of agile adoption and use.

In the next section, we review empirical studies that have examined the adoption and use of software development methodologies. We build on this extant literature to develop our research model and hypotheses in section three, which also describes our data collection and analyses procedures. In Section 4 we discuss our key findings and their implications, and in Section 5 we acknowledge the study's limitations and offer some suggestions for future work. Finally, we summarize our results and its relevance for research and practice in the concluding section.

2. Review of prior empirical research

Prior empirical research has drawn on multiple frameworks including the theories/models of reasoned action [7], planned behavior [8], innovation diffusion [9], and technology acceptance [10,11] to better understand the adoption and use of methodologies, practices, processes and tools in software development. We elaborate on some of the key studies within this research stream that are summarized in Table 1. Leonard-Barton [12] found that in addition to training and perception of specific benefits, supervisors, influential peers and clients also had an impact on use. The importance of perceived advantages and subjective norm is also reflected in Riemenschneider et al.'s [13] study, which concluded that organizational mandate, compatibility or fit with how developers worked, and subjective norms were important predictors. More recently, Kaemar et al. [15] found that perceptions about the quality of outputs attributable to the use of a development methodology was positively associated with perceived usefulness, while perceptions about the challenges to using a development methodology was negatively related to perceived ease of use.

The relevance of organizational factors has been highlighted in Roberts et al. [17] study which concluded that transitional support from organizations, functional management involvement/support and external support from consultants were important in the

implementation of software development methodologies. Similarly, Higgins and Hogan [18] found that perceived success of Information Engineering implementation was influenced by cross-functional team spirit, user participation and top management support. Khalifa and Verner [20] reported that the use of prototyping and waterfall methodologies was not only influenced by developers' perceptions of their impact on process quality but also by facilitating conditions such as team size and organizational support. Iivari and Huisman [21] concluded that organizational culture played a role in the adoption of software development methodologies.

Some studies have also examined the diffusion of specific software development processes and tools. For example, Green and Hevner [22] and Green et al. [23] examined the diffusion of Personal Software Process, a software process innovation. They found that perceived control had a positive impact on the diffusion of the innovation. Hardgrave and Johnson [24] also found support for the role of perceived behavioral control in systems developers' intentions to use object-oriented (OO) systems development. However, in their study, its effect was mediated by perceived organizational usefulness.

Besides innovation-related factors, some adopter characteristics have also been identified as relevant to the acceptance of innovations in software development. For example, Kozar [26] found that the adopters of a novel technique for requirements definition tended to be younger with less tenure at their organizations.

Several studies have examined the adoption of computer-assisted software engineering (CASE) tools. For example, Chau [30] studied the acceptance of CASE tools and found that implementation gap (i.e., the gap between the adopter's current skills and knowledge and those required by the new technology) had a negative effect on the adopter's perceived ease of use and near-term usefulness of adoption, while transitional support had a positive influence on ease of use. Similar results about the importance of training are reported by Rai and Patnayakuni [31] and Dietrich et al. [32]. Iivari [33] investigated the adoption of CASE tools and found that relative advantage, voluntariness and management support were strong predictors of its use. The significance of managerial-backing in CASE adoption has been echoed in other studies including Rai and Patnayakuni [31] and Morris et al. [34].

Similar to CASE tools, interest in agile development has generated a number of empirical studies in the last 10 years [37–43]. However, most of this research has focused on examining the benefits and limitations of agile use. While some studies have reported the experiences of using agile at a single organization (e.g., [37]), others have used experimental settings to evaluate its advantages and drawbacks (e.g., [40]). There is far less research on the adoption and use of agile. For example, Salo and Abrahamsson [44] found that about half of the projects they examined used XP at least sometimes, and about a quarter of them used Scrum at least to some extent. Not surprisingly, they also found that respondents who had experience with XP and/or Scrum also had more favorable opinions about their usefulness. These results are echoed by Laanti et al. [45], who discovered that there was a positive association between length of agile experience and attitudes towards its usefulness. Tolfo and Wazlawick [46] argued that an organization's cultural orientation with respect to innovation and risk, detail, outcome, people, team, aggressiveness, and stability could have both positive and negative implications for XP's adoption. Drobka et al. [47] also emphasized the role of organizational culture in addition to training and coaching.

Although there is considerable diversity in the research approach, levels of analysis and targeted innovation, there is also convergence on some key factors that are instrumental in the adoption of and use of software development methodologies, practices, processes and tools. Using the conceptual framework

Table 1

Summary of prior empirical research on the adoption and use of software development methods/practices/processes/tools.

Study	Method	Determinants of adoption and use
Leonard-Barton [12]	Structured methodology	Training, benefits, supervisors, influential peers and clients
Riemenschneider et al. [13]	Customized structured methodology	Benefits, organizational mandate, work compatibility and subjective norms
Hardgrave et al. [14]	Customized structured methodology	Usefulness, social pressure, compatibility and organizational mandate
Kaemar et al. [15]	Any formal methodology	Usefulness and challenges
Zmud [16]	Six software development practices	Top management's attitude and openness to change
Roberts et al. [17]	Custom methodologies	Support from organization, functional management and external consultants
Higgins and Hogan [18]	Information Engineering	Cross-functional spirit, user participation and top management support
Rahim et al. [19]	Structured methodology	Organizational type and maturity level
Khalifa and Verner [20]	Prototyping and waterfall	Benefits, team size and organizational support
Iivari and Huisman [21]	Any formal methodology	Organizational culture
Green and Hevner [22] and Green et al. [23]	Personal Software Process	Perceived control, voluntariness, champion support, training and degree of novelty
Hardgrave and Johnson [24]	OO development	Perceived control and organizational usefulness
Grossman et al. [25]	UML	Understandability, accuracy, consistency and flexibility
Kozar [26]	Requirements definition technique	Age, attachment to current method, and risk averseness
Sultan and Chan [27]	OO technology	Teamwork, leadership, communication and risk attitudes
Sherif and Vinze [28] and Sherif et al. [29]	Software reuse	Defined reuse policy, funding, education/training, software repository, reuse champion and infrastructure to create reuse routines
Chau [30]	CASE	Knowledge/skills, transitional support, and ease of use
Rai and Patnayakuni [31]	CASE	Training and management support
Dietrich et al. [32]	CASE	Knowledge of tools, flexibility of tools, organizational willingness and ability to change
Iivari [33]	CASE	Relative advantage, voluntariness and management support
Morris et al. [34]	CASE	CASE champions, IT department stability, external sources for CASE knowledge, job rotation, and performance pressures
Nelson and Rottman [35]	CASE	Pressure to address performance gaps
Orlikowski [36]	CASE	Intentions and actions of key players and the social context of systems development and implementation processes
Talfo and Wazlawick [46]	XP	Organizational culture
Drobka [47]	XP	Organizational culture, training and coaching

proposed by Chan and Thong [48], these factors can be classified as relating to ability, motivation and innovation characteristics. Among ability-related factors, training and self-efficacy are dominant predictors, while organizational culture, receptiveness to change, the presence of innovation champions/supporters and subjective norm are critical motivational factors. Finally, almost every study has shown the criticality of innovation-related factors including perceptions about its usefulness, ease of use, compatibility and flexibility.

The extant empirical literature provides significant insights into the drivers of software development innovations. We draw on this research stream and extend it to the study of agile adoption and use.

3. Research design and conduct

3.1. Research model and background

Fishbein and Ajzen's [7] theory of reasoned action (TRA) models an individual's intention to enact a behavior as a function of his/her own beliefs about engaging in that behavior, as well as the influence of other people who are important to the individual (subjective norm). TRA and other intention-based theories/models including the theory of planned behavior and the technology acceptance model (with its parsimonious set of just two salient beliefs – usefulness and ease of use) have been the foundation of many empirical studies on user acceptance of information technologies. Of course, as illustrated in the previous section, these theories have also provided the framework for exploring the adoption and use of innovations in software development.

Our research model (Fig. 1) reflects the importance of TRA in information technology (IT) use research, and incorporates salient

beliefs about the perceived benefits and limitations of agile methodologies and the influence of important others (subjective norm) as predictors of its use. In addition, our model also includes training as its pertinence to the adoption of software development innovations has been repeatedly validated in prior research. While enabling factors such as perceived usefulness and organizational support are typically examined in the context of innovation adoption, few studies have focused on detracting barriers/hindrances whose influence may be just as relevant. We address this deficiency in the current literature by explicitly including perceived hindrances in our model. Further, drawing on the dialectic perspective on innovation, we incorporate an interaction effect between perceived benefits and perceived hindrances. Finally, our model includes two control variables to account for differences in the size of the respondents' organizations and their software development experience.

The dialectic method seeks to resolve conflicts and advance knowledge through the critical exchange and examination of arguments and counter-arguments. As a process for finding resolution between different points-of-view through dialog, its emergence in Western philosophies dates back to the Socratic period. More recently, information systems researchers have embraced Hegelian dialectics, which posits that development is the result of achieving synthesis or integration between a projected thesis and its negation or anti-thesis [49], as a theoretical lens for examining organizational change processes in the development and implementation of information systems.

For example, Sabherwal and Newman [50] applied dialectic theory to understand the systems development process which can be characterized by both a desire to maintain continuity or persistence as well as pressures to yield or accommodate changes. In this study persistence was equated to the prevailing thesis or the current set of assumptions, values and beliefs (T1), and change to

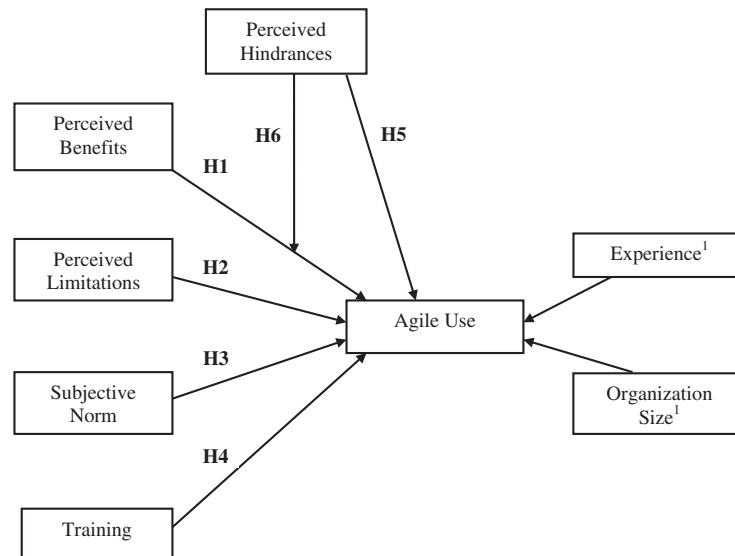


Fig. 1. Research model and hypotheses. 1 – Control variables.

either an antithesis or competing set of assumptions, beliefs, and values (not T1) or a synthesis (T2) – a new integrated set of assumptions, values and beliefs that may have elements of T1 and not T1 but yet was distinct from the two competing theses.

Sabherwal and Newman [50] used dialectical theory to interpret the sequence of events and the interactions between content (e.g., available IT solution) and process elements (e.g., project leadership) that occurred in systems development projects at three different organizations. They observed that persistence and change can co-exist in a systems development project, and the latter could be triggered by organizational learning, technological progress, problems related to performance or by an antithesis embedded within a new solution (synthesis) that resolved a previous contradiction. Further, they concluded that a dialectical view of systems development, rather than providing simple causal explanations, offered a complex, insightful and emergent perspective that explicitly recognized the social construction of technology by organizational actors.

Dialectical theory has also been used to explain the contradictions that arise when systems are implemented in organizations. For example, Pawlowski et al. [51] used a dialectic approach to analyze the contradictory impacts of a Systems, Applications, and Products in Data Processing (SAP) implementation at a large manufacturing and distribution company. They found that while the adoption of the new enterprise-wide system was less flexible and less amenable to organizational control than the independent systems it replaced, it also provided the employees a more global view of the organization and an opportunity to be more innovative and empowered in their decision-making.

In their analysis of enterprise resource planning (ERP) implementation at 13 North American manufacturing companies, Robey et al. [52] found support for the dialectics of organizational learning [53]. They observed that the incongruities between the ingrained knowledge of legacy systems' existing processes and the new processes demanded by the ERP system created daunting obstacles to acquiring knowledge and learning to use the newly implemented system. Soh et al. [54] also examined ERP implementation, but at a large hospital in Singapore. They viewed the misalignments created by the implementation as a function of differences in the structural features of the implemented system and the structural needs of the organization that are manifested as opposing or dialectic forces between integration versus differentiation, process- versus

functional-orientation, flexibility versus restrictiveness and packaged solutions versus unique requirements.

As theorized in conceptual works (e.g., [55]) and demonstrated in empirical research (e.g., [56]), a dialectical perspective can be employed not only to enhance our understanding of tensions and contradictions in the adoption of innovations in organizations, but also provide insights into their management and resolution. In the context of agile adoption, we can equate perceptions about the benefits of this new methodology as the antithesis or not T1 seeking to replace the prevailing thesis or T1 articulated as the desire to maintain the status quo or hindrances. Our research model attempts to capture the influence of these dueling forces of T1 and not T1 and the effects of their interaction which represents their integrative synthesis or T2 on agile adoption.

3.2. Variables and hypotheses

During the last decade a few development methods/processes and project management techniques including eXtreme Programming (XP), Agile unified process (AUP) and SCRUM that incorporate specific practices reflecting agile principles have emerged. Recent studies have highlighted XP and its related practices including pair programming as the most widely used method/technique [40,57,58]. In this study, we are interested in the level of agile use, and define it as the *extent to which specific agile development methods and techniques (e.g., XP, AUP, pair programming) are used*.

Theories that explain the adoption and use of an innovation emphasize the importance of beliefs and perceptions about its benefits and limitations including its usefulness, ease of use, compatibility, trialability and visibility. Empirical work has also established the prominent role of innovation-related attributes in the assimilation of new software development methods/tools/techniques. Agile adoption may place considerable demands on software developers to not only learn new tools and techniques, but also require them to make significant changes to their current work patterns, processes and schedules. Clearly, software developers have to perceive specific benefits in adopting agile to make the transition from existing methodologies. We define perceived benefits as the *extent to which agile use is perceived to improve software development processes and outcomes (e.g., improved quality, faster delivery, reduced costs)*, and expect it to have a positive association with agile use.

H1. There is a positive association between perceived benefits and agile use.

Potential adopters may perceive both benefits and limitations associated with the target innovation. These perceptions are especially pertinent in the case of an innovation which is considered as a possible replacement to the existing system, method or process. It can be expected that the innovation will be evaluated on its relative advantages as well as its inherent weaknesses. We define perceived limitations as the *extent to which agile use is perceived to adversely affect software development processes and outcomes (e.g., poor documentation, more bugs/defects, lack of standards)*, and expect it to be negatively related to agile use.

H2. There is a negative association between perceived limitations and agile use.

Intention-based theories explicitly recognize the influence of important others in an individual's intent to engage in a certain behavior. There is also extensive empirical evidence that validates the theorized role of subjective norm in the prediction of innovation use among individuals [12–14]. In deciding to use agile, software developers can be expected to be motivated by others who are important to them and whose opinions they value. We define subjective norm as the *extent to which important referent groups/individuals/organizations influence agile adoption and use*.

H3. There is a positive association between subjective norm and agile use.

Empirical research on the acceptance of software development innovations has consistently demonstrated the importance of training [31,32]. Further, it can be argued that self-efficacy, which has also been shown to influence the adoption of these innovations, may receive a boost through training. Since agile adoption requires adherence to different principles and the use of new processes, techniques and practices, training can be expected to play a vital role in encouraging its use. We define training as the *amount of agile education received by potential adopters*.

H4. There is a positive association between training and agile use.

Framed within the context of dialectical theory, the introduction of an innovation can be equated to an antithesis (not T1) competing to replace the prevailing thesis (T1). The desire to maintain T1 or the status quo may trigger self-preservation measures which, from the perspective of not T1's proponents, are manifested in the form of road-blocks or hindrances. For our study, we define perceived hindrances as the *extent to which there are impediments or barriers (e.g., lack of peer and institutional support) to agile use*.

H5. There is a negative association between perceived hindrances and agile use.

From a dialectical perspective, change is realized by the interplay between T1 and not T1. Extending this concept to innovations, we can expect the adoption process to be characterized by interactions between arguments for the change and counter-arguments for maintaining the status quo. Further, it can be argued that the prominence and relevance of the arguments for not T1 and counter-arguments for T1 are functionally inter-dependent. In other words, the relevance of the arguments for not T1 will be directly proportional to the strength of the arguments for T1. Therefore, while perceived hindrances (counter-arguments for T1) can adversely impact agile use, it can also elevate the importance of perceived benefits (arguments for not T1). We model this relationship between perceived benefits and hindrances as an interaction effect in our final hypothesis.

H6. The relationship between perceived benefits and agile use is moderated by perceived hindrances.

3.3. Data collection

Data for this study were collected through an anonymous online survey of software development professionals who are most likely to be adopters of agile development. We identified 15 Yahoo online discussion groups that focused on agile software development and after obtaining permission from the groups' moderators, posted a solicitation message inviting members who had experience using agile development approaches to complete an anonymous web-based survey. The survey asked respondents to provide their perceptions about the benefits, limitations and hindrances associated with agile software development. They were not instructed to confine their responses to specific types of projects (e.g., collocated versus distributed; new versus maintenance).

Each of the research constructs was measured with multiple item scales. Perceived benefits, perceived limitations and perceived hindrances were measured with six, eight and four items respectively. Survey participants had to specify the extent to which each of the items, adapted from [12,13,15,20,33], had an impact on their agile adoption decision. Subjective norm was assessed by asking respondents to indicate the extent to which four referent groups, identified from [14,34,36], influenced their decision to adopt agile methods. Training in agile methods was gauged by asking respondents to state the amount of time they had spent on different forms of training [22,28,31,47]. Agile use was measured by asking respondents to indicate the extent to which four specific agile processes and methods were employed in software development projects [2,6]. Finally, experience, one of the two control variables, was captured by asking the respondents their years of software development experience, while organizational size, the other variable, was determined through three items – annual revenue, number of employees and number of software development professionals.

We received 98 responses from software professionals who had an average of 15.5 (median = 15.0) years of experience with software development and 3.9 (median = 3.0) years of agile experience. The majority of the respondents (81) identified their country of residence as the United States of America followed by Canada, India and the United Kingdom with 4 respondents each and Australia, Botswana, Colombia, Mexico and New Zealand with 1 respondent each. While the respondents had a variety of job titles, including some unique titles such as Agile/XP Coach and Chief Agilist, the most popular categories were software developer/architect/engineer/programmer (48%), director/manager/leader (34%), and analysts/consultants (11%).

The respondents' organizations covered all the major industry sectors, but most were from the IT (31%) and Finance/Insurance/Real Estate (16%) sectors. The median annual organizational revenues category was US\$10 million to US\$49 million; the median number of employees in the organizations was 100–199; and the median number of software professionals in the organizations was 20–49. (Note: respondents were asked to specify a range rather than provide a specific number for each of these three measures.)

Our study solicited responses from individuals who were knowledgeable and/or interested in agile development. Therefore, our sample pool of respondents was more representative of likely adopters of agile and not of all developers.

3.4. Data analysis

We used partial least squares (PLS) regression to statistically analyze the research model and hypotheses. PLS integrates features of principal components analysis and multiple regression to test predictive models that relate dependent and independent variables. One of the main advantages of PLS and LISREL (linear structural relations), another structural equation modeling (SEM)

technique, over first generation statistical tools (e.g., factor analysis, regression) is the ability to combine the empirical assessment of both the measurement and structural models [59–62]. In SEM, the measurement model refers to the linkages between the research constructs (latent variables) and their indicators (manifest variables) and the structural model captures the hypothesized causal relationships among the research constructs. For example, in our study, the *measurement model* included the relationships between the latent construct – perceived benefits – and each of its six indicators (please refer to [Appendix A](#)) and the *structural model* included the relationship between perceived benefits and agile use (please refer to [Fig. 1](#)).

For this study, SmartPLS 2.0.M3 software [63] was used to conduct the analysis. PLS was chosen for two reasons. First, this SEM technique is recommended when the sample size is relatively small. A second reason for choosing PLS is that this research is one of the first attempts to explain agile use among its adopters, and is, therefore, in the preliminary stages of theory development and more exploratory in nature. In general, PLS is considered to be more appropriate for testing research models that are relatively new and in a nascent phase [64].

3.4.1. Measurement model

In assessing the measurement model, the loadings of the measurement items on their respective constructs are relevant [65,66]. Based on Chin's [67] suggestion, we included only those items that had loadings of 0.707 or greater ([Table 2](#)). Chin [67] also recommends that internal composite reliability (ICR) scores be used to assess the reliability of reflective constructs. These scores are included in [Table 1](#), and are all above the 0.70 level suggested by Fornell and Larcker [68].

Convergent and discriminant validities of the measurement items were evaluated by comparing (a) each item's loading on its construct to that of its cross-loadings, and (b) the square root of the average variance extracted (AVE) of each construct to its correlations with other constructs. As shown in [Table 3](#), the loadings for

each item on its assigned construct are higher than its cross-loadings on the other constructs [61]. Further verification of convergent and discriminant validities was conducted by comparing the square root of the AVE of each construct to its correlations with the other constructs. [Table 4](#) reveals that the square root of the AVE of each construct is higher than its correlations with the other constructs [67,69].

3.4.2. Structural model

The study hypotheses were tested by examining the results for the structural models generated by SmartPLS 2.0.M3. Similar to regression analysis, PLS produces squared multiple correlations (R^2) for the endogenous construct in the model that could be used to assess its explanatory power. Since we wished to test the moderating effect of perceived hindrance on the relationship between perceived benefits and agile use, we examined two structural models – (a) a main effects model and (b) an interaction model. Results from these two models were then compared to evaluate the size of the moderating effect [70].

PLS generates the path coefficients for the relationships modeled among the constructs. The significance of these coefficients was assessed using the bootstrap procedure (with 500 sub-samples) that provided the t -values for each path estimate. [Figs. 2 and 3](#) show the results for the main effects model and interaction model respectively. The main effects model reveals significant relationships between agile use and subjective norm (H3) and training (H4). In addition, the two control variables – software development experience and organizational size exhibit significant negative relationships with agile use. However, there is no support for the posited relationships between agile use and perceived benefits (H1), perceived limitations (H2), and perceived hindrances (H5).

For the interaction model, we introduced an interaction construct by using the product indicator approach as suggested by Chin et al. [70]. With this approach, the interaction construct (i.e., perceived benefits * perceived hindrances) was reflected by all possible products of the standardized items used to indicate the predictor (i.e., perceived benefits) and moderator (i.e., perceived hindrances) constructs. All twelve items (the four items for perceived benefits * the three items for perceived hindrances) that reflected the interaction construct had loadings above 0.707 ranging from 0.814 to 0.981. Further, the reliability scores and square root of the average variance extracted for this construct were also above the recommended minimums (Cronbach's alpha: 0.98; composite reliability: 0.98; square root of AVE: 0.89).

The interaction model shows that the interaction construct (perceived benefits * perceived hindrances) is significant and it boosts the influence of perceived benefits on agile use by 0.25. With the strong interaction effect, the influences of subjective norm and training on agile use are slightly reduced; but the relationships remain significant and continue to offer support for hypotheses H3 and H4. However, the control variables no longer have significant effects on agile use in the interaction model.

In order to assess the contribution of the interaction term, we computed Cohen's effect size f^2 measure as follows [70, pages 26]:

$$f^2 = [R^2(\text{interaction model}) - R^2(\text{main effects})] / R^2(\text{interaction model})$$

Based on Cohen and Cohen [71], the effect size f^2 of 0.22 is between a medium (0.15) and high effect (0.35). The magnitude of this effect size offers support for the interaction model and H6, suggesting that perceived benefits play a significant role in influencing adopters' use of agile processes and methods if they perceive hindrances within their organizations to their use. In other words, the importance of perceived benefits as a predictor of agile

Table 2
Loadings and reliabilities of measurement model.

Construct	Item ^a	Loading	Composite reliability	Cronbach's alpha
Perceived benefits	PB1	0.864	0.89	0.87
	PB3	0.724		
	PB5	0.921		
	PB6	0.764		
Perceived limitations	PL1	0.780	0.91	0.89
	PL2	0.712		
	PL3	0.752		
	PL4	0.880		
	PL6	0.773		
	PL7	0.764		
	PL8	0.708		
Perceived hindrances	PH2	0.853	0.87	0.78
	PH3	0.883		
	PH4	0.753		
Subjective norm	SN2	0.862	0.87	0.78
	SN3	0.861		
	SN4	0.767		
Training	TR2	0.932	0.89	0.77
	TR3	0.865		
Organization size	OS1	0.907	0.96	0.93
	OS2	0.956		
	OS3	0.955		
Agile use	AU1	0.740	0.87	0.79
	AU3	0.883		
	AU4	0.876		

^a The complete wording for the items are provided in [Appendix A](#).

Table 3

Loadings and cross-loadings of measurement items.

Construct	Item ^a	PB	PL	PH	SN	TR	OS	EX	AU
Perceived benefits (PB)	PB1	.864	.016	.377	.083	.352	.106	.309	.154
	PB3	.724	–.089	.274	.094	.142	.022	.297	.016
	PB5	.921	–.049	.324	.036	.266	–.054	.141	.222
	PB6	.764	–.076	.259	–.002	.181	.118	.335	.064
Perceived limitations (PL)	PL1	–.020	.780	.315	.238	.169	.214	.021	–.094
	PL2	.009	.712	.295	.277	.131	.186	–.020	–.051
	PL3	–.001	.752	.102	.200	.024	.107	–.067	–.112
	PL4	.003	.880	.279	.174	.051	.162	–.069	–.082
	PL6	.037	.773	.279	.184	.174	.142	–.060	–.078
	PL7	.010	.764	.317	.225	.089	.153	.071	–.065
	PL8	–.119	.708	.223	–.043	.057	.079	–.063	–.173
Perceived hindrances (PH)	PH1	.242	.233	.853	.006	.209	.279	.096	.069
	PH3	.333	.197	.883	.106	.275	.132	–.014	.148
	PH4	.344	.394	.753	.101	.296	.123	.120	.103
Subjective norm (SN)	SN2	.116	.184	.075	.862	.202	.251	–.150	.261
	SN3	–.005	.146	.123	.861	.255	.190	.072	.191
	SN4	–.003	.134	.048	.767	.145	.139	.005	.165
Training	TR2	.305	.159	.314	.248	.932	.104	–.023	.362
	TR3	.266	.034	.256	.180	.865	.095	–.064	.260
Organization size (OS)	OS1	.010	.160	.156	.272	.092	.907	–.044	–.094
	OS2	.016	.171	.202	.222	.084	.956	–.035	–.125
	OS3	.053	.174	.181	.204	.131	.955	–.069	–.139
Experience (EX)	EX1	.264	–.046	.064	–.047	–.044	–.054	1.000	–.151
Agile use (AU)	AU1	.130	–.180	.075	.039	.242	–.044	–.067	.740
	AU3	.140	–.050	.089	.290	.384	–.058	–.173	.883
	AU4	.209	–.152	.170	.251	.252	–.197	–.120	.876

^a The complete wording of the items are provided in [Appendix A](#).**Table 4**Square root of AVE and correlations among constructs.^a

AVE/correlation	PB	PL	PH	SN	TR	OS	EX	AU
Perceived benefits (PB)	.82							
Perceived limitations (PL)	–.04	.77						
Perceived hindrances (PH)	.38	.32	.83					
Subjective norm (SN)	.06	.19	.10	.83				
Training (TR)	.32	.12	.32	.24	.90			
Organization size (OS)	.03	.18	.19	.24	.11	.94		
Experience (EX)	.26	–.05	.06	–.05	–.04	–.05	1.00	
Agile use (AU)	.19	–.14	.14	.26	.35	–.13	–.15	.84

^a The square root of AVE is shown as the diagonal elements; while the off-diagonal elements are the correlations.

use is contingent on perceived hindrances. We have summarized the results in [Table 5](#), and discuss its implications in the next section.

4. Key findings and implications

The results of this study suggest that the drivers of agile software development use are contingent on perceived hindrances. If there are no perceived hindrances to the use of agile methods, then (a) subjective norm and training are strongly and positively related to agile use, (b) software development experience and organizational size are moderately and negatively related to agile use, and (c) perceived benefits and perceived limitations are unrelated to agile use.

On the other hand, if there are perceived hindrances to the use of agile methods, then there is a different combination of factors that influence agile use. Specifically, perceived benefit moderated by perceived hindrances is strongly and positively related to agile use. In other words, it is the interaction between perceived benefits and perceived hindrances that is most important in agile use. Further, subjective norm and training are still positively related to

agile use, though their effects are weaker, and finally, software development experience and organizational size are unrelated to agile use.

These results are quite interesting, and have useful implications for practice. In the first case, while in most organizations it is unlikely that there will not be any perceived hindrances, it is important to realize that in such instances it is not the perceived benefits or limitations of agile that are critical. Rather, agile use will depend on the presence of support and approval from key people (subjective norm) and the availability of relevant training to facilitate the transition. Further, the more experience the developers have and the larger the organization, it is less likely that agile will be used.

Awareness of these factors can guide management in their choice of appropriate implementation strategies, and allocation of time, money and resources in supporting a transition to the use of agile methods. While it would be important for management to come to a realistic understanding of whether the organization perceives hindrances to agile use, if it appears that there is no opposition (i.e., lack of support to continue T1), then it will be imperative to identify and publicize the key influential people within the organization who use or support the use of agile

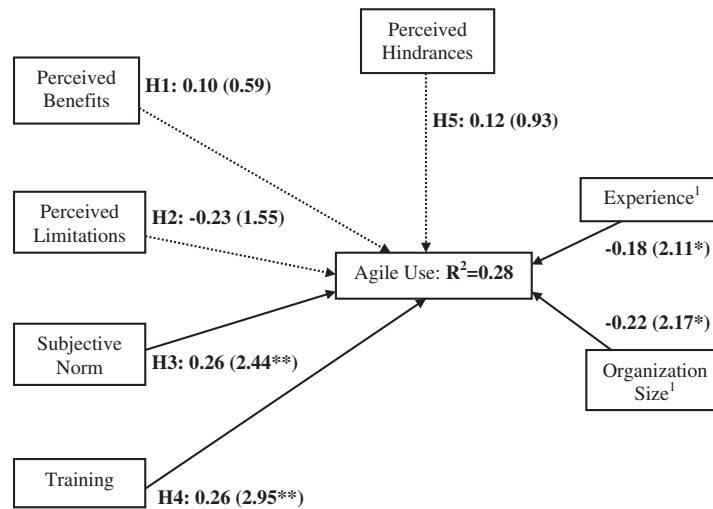


Fig. 2. Structural model results – main effects model. 1 – Control variables. T-values are shown in parentheses. Insignificant paths are shown with dashed lines. * $p < 0.02$, ** $p < 0.01$.

methods. Likewise it will be important to identify and provide useful training on agile methods. Finally, management should recognize that the larger their organization and the more experienced their developers, the more resistance there may be to a transition to agile, and thus they will need to be especially patient and take the time to listen to and work with their developers in the transition process.

In the second case, if there are perceived hindrances, then it is the interaction between perceived benefits and perceived hindrances that is most important in agile use. In other words, it becomes necessary for management to publicize and promote the perceived benefits of agile (not T1) to overcome the arguments for maintaining the status quo (T1). This may also require management to listen to developers' views and concerns and forge an integrative synthesis (T2). For example, management may rely on pilot and proof of concept projects to ease the transition to agile or promote hybrid development methods which harness the strengths of both agile and plan-based processes and techniques.

The importance of the interaction between perceived benefits and hindrances suggests that adopting a dialectical perspective can be critical to the acceptance and diffusion of innovations in

software development. A dialog about the perceived benefits and hindrances may be just what is necessary in order to reach consensus on a proposed new software development approach. Further, this dialog may be instrumental in resolving disagreements and paving the way for an acceptable software development approach to emerge.

It is interesting to observe that perceived benefits are only a strong factor in determining agile use when they exist in an environment where there are also perceived hindrances. When perceived benefits exist in an environment where there are no (or few) perceived hindrances, then other factors dominate, and the benefits are not the driving force. This phenomenon supports the dialectical perspective of contradiction and change in which there could be a continual dialog, and even a tension, between various actors or ideas, and it is in the resolution of this tension that solutions are found. The fact that perceived benefits (not T1) are only a strong determinant when seen in the context of perceived hindrances (T1) offers a compelling argument for incorporating dialectic theories to study the diffusion of software development innovations.

While it is important for managers to be aware of the existence of perceived hindrances or not, and the implications for practice

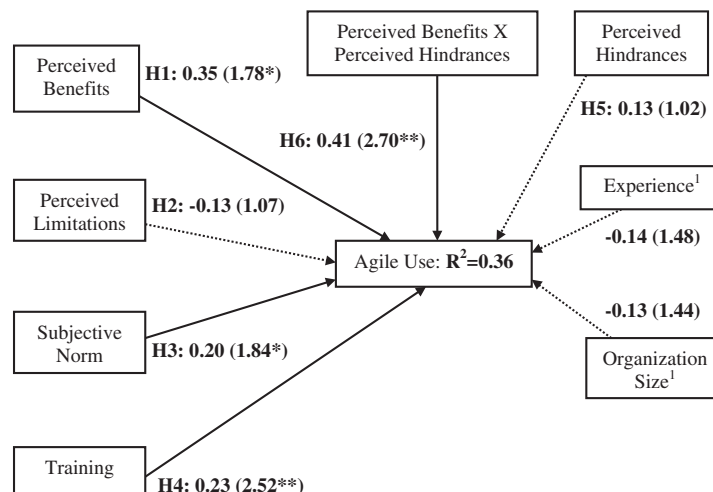


Fig. 3. Structural model results – interaction effect model. 1 – Control variables. T-values are shown in parentheses. Insignificant paths are shown with dashed lines. * $p < 0.05$, ** $p < 0.01$.

Table 5
Summary of results.

Hypothesis	Statistical analysis – interaction model	Supported?
H1: There is a positive association between perceived benefits and agile use	Significant path coefficient of 0.35	Yes
H2: There is a negative association between perceived limitations and agile use	Non-significant path coefficient of –0.13	No
H3: There is a positive association between subjective norm and agile use	Significant path coefficient of 0.20	Yes
H4: There is a positive association between training and agile use	Significant path coefficient of 0.23	Yes
H5: There is a negative association between perceived hindrances and agile use	Non-significant path coefficient of 0.13	No
H6: The relationship between perceived benefits and agile use is moderated by perceived hindrances	Significant path coefficient of 0.41 Significant effect size of 0.22 between the main effects and interaction models	Yes

discussed above that derive directly from this study's research, some additional issues may be relevant as well. First, if the members of the organization perceive hindrances to agile adoption, then it is especially important to carefully manage the transition [72]. Transitions often pose unique challenges, but for many organizations it is the failure to aptly handle the hindrances to change that inevitably derails their implementation efforts.

Second, as described above, when this tension between perceived benefits and perceived hindrances does not exist, organizational size, software development experience, subjective norm and training are the dominant factors influencing the use of agile development approaches. This makes sense because, unless people see a reason to do something other than what they are familiar with and experienced at, they will probably continue to do those same things. Further, in large organizations, change is harder to facilitate than in smaller organizations, because of organizational momentum and culture.

This implies that smaller organizations may be able to more readily apply new technologies and ideas, such as agile development, because they are not constrained by organizational history, culture, and momentum. It may also suggest that it is important for champions of new approaches to identify possible hindrances to the new approaches, as this identification of possible hindrances may actually help potential adopters to look more closely at the perceived benefits, and to take these possible benefits more strongly into account as they decide whether to use or not use the new approaches.

These findings corroborate earlier studies (e.g., [13,14,19,20,26,28,31,32]) which have identified the importance of clear benefits, subjective norm, organizational size, developer experience, organizational barriers, and training provided in the adoption of a software development approach. However, the findings of this study go further and indicate that the relationship between perceived benefits and hindrances is even more important than these other factors on their own. In fact, when hindrances are perceived to exist, the importance of the other factors becomes secondary to its interaction with perceived benefits. Thus, a dialectical approach may be helpful in determining what approach to adopt, and in actually making the adoption successful.

While some of the earlier research focused variously on structured (traditional) methods, perceived usefulness, and compatibility with developers' ways of working, our research focused solely on the adoption and use of agile methods and the importance of perceived benefits and perceived hindrances. Even so, in many cases, our research corroborates earlier findings in that both indicate the importance of perceived benefits, subjective norm, training, and support of management.

5. Limitations and suggestions for future research

As is the case with most field research, our study has some limitations. First, the sample size is only 98 respondents. While this is a reasonably good size, it would be better to have a larger sample.

Second, a larger study may also help increase the diversity of the sample, and thus the ability to generalize to a wider population. This study only drew from people who had signed up for and were currently active on a set of Yahoo groups that discussed agile topics. Therefore, our sample is not representative of the broader population of developers world-wide. Third, our respondents' level of software development experience (median of 15 years of overall experience and a median of 3 years of agile experience) may not be representative of the population of developers.

Fourth, our data collection strategy introduced validity threats to the extent that we could not verify that every one of our survey responses originated from a member of our target population and that there were no duplicate submissions. We tried to mitigate this problem by posting our survey to targeted discussion groups that were moderated and active. We used a validation feature of the survey development tool to prevent a respondent accidentally (or, impatiently) submitting the survey twice. Further, we manually checked each submission to ensure that there were no duplicates. Since our survey was openly available online, we cannot be absolutely certain that every response was valid. However, given the targeted nature of our solicitation, the length of the questionnaire, the domain-specific knowledge required to answer the questions, and the detailed open-ended answers we received from about 40% of our respondents, we are confident that the risk of invalid data is minimal.

Finally, our study is quantitative and relies on scaled questionnaire items for measuring the research constructs. While we have used multiple items in the measurement of each construct and rigorous statistical tests to assess their reliabilities and validities, additional data in the form of qualitative insights could help corroborate our research findings. Future studies can address these limitations by gathering data through multiple methods from a larger and broader sample of developers representing greater diversity in agile use and software development experience.

While our research has provided some insights into the factors that drive agile use, there are still a number of questions related to the value and appropriateness of agile approaches that can be addressed by future studies. Some of these questions include: (1) Is there an agility continuum? If so, what is the appropriate level of agility for any given software development project? (2) What observable and measurable benefits (time, cost, quality), if any, can we expect from adopting more agile approaches? (3) In what situations is it appropriate to adopt more agile approaches, and in what situations is it more appropriate to adopt more prescribed, less agile, ones? (4) How can we effectively migrate to the use of more agile development methodologies? (5) Can agile approaches, where it is usually expected that a customer is on-site with the developer much of the time, be employed effectively in distributed environments? (6) Could virtual reality environments be used to help support such distributed situations? Providing empirical answers to these questions would be valuable in advancing the state of knowledge and application in software development.

Table 6
Key findings and implications.

	Perceived hindrances	
	No	Yes
Findings	- Perceived benefits and limitations are unrelated to agile use - Subjective norm and training are strongly and positively related to agile use - Software development experience and organizational size are moderately and negatively related to agile use	- Perceived benefit moderated by perceived hindrances (the interaction between the two) is strongly and positively related to agile use - Subjective norm and training are strongly (though weaker than when there are no perceived hindrances) and positively related to agile use - Software development experience and organizational size are unrelated to agile use
Implications for practice	- Perform training on agile use - Find and publicize influential individuals and entities who have adopted agile and/or recommend its use - Agile is more likely to be adopted in smaller organizations - Agile is more likely to be adopted by novice developers	- Perform training on agile use - Find and publicize influential individuals and entities who have adopted agile and/or recommend its use - Agile is likely to be adopted by organizations of all sizes - Agile is likely to be adopted by both novice and experienced developers
Implications for research	- Study best ways to train on use of agile methods - Study best ways for influential individuals and entities to champion agile use - Study how to make less experienced developers more adept at using agile methods, and how to make up for their relative lack of experience	- Study best ways to train on use of agile methods - Study best ways for influential individuals and entities to champion agile use - Study how to leverage experienced developers, and how to transfer their experience/skills to the less experienced developers - Study how to effectively elicit perceived benefits from developers - Study how to best acknowledge perceived hindrances, yet at the same time focus on perceived benefits - Study how to have a dialog about the two, how to facilitate and embrace this dialectic

6. Conclusions

Agile software development has emerged as an important alternative to traditional plan-based approaches. While some surveys have attempted to gauge the extent of agile use, our goal was to examine the drivers of agile use. Our results show that enabling factors such as subjective norm and training, that are typically important in the context of software development innovations, are also relevant in the adoption of agile processes and methods. Further, the incorporation of a dialectic perspective provides insights into the role of detracting factors and their interactions with enablers. Specifically, our analysis shows that the importance of perceived benefits is conditional on the presence of perceived hindrances.

These findings, summarized along with their implications in Table 6, highlight the need for adoption researchers to examine both enabling and detracting factors and the dialectic interplay between the two. In particular, since training, subjective norm, experience, and the interaction between perceived benefits and perceived hindrances are the key factors, research might focus on how to leverage these factors effectively, and how to facilitate the dialog regarding perceived benefits and hindrances.

For practitioners, the results suggest a more subtle and contingent approach to managing the adoption of innovations such as agile development. While there are some general recommendations, others are dependent on whether there are perceived hindrances or not. In all cases, training and subjective norm are important. In cases where there are perceived hindrances, it is important to elicit and publicize both the perceived benefits and hindrances, and to allow a healthy dialog about the interplay between the two. The dialog and dialectic interplay among and between developers and managers regarding the pros and cons of any

proposed new development approach is fundamental for an integrative synthesis to emerge and be accepted.

Appendix A. Survey items

Perceived benefits (1 – not at all; 4 – to some extent; 7 – to a great extent)

- PB1 – Improved software quality
- PB2 – Faster time to delivery^a
- PB3 – Lower development costs
- PB4 – Better meets customer needs^a
- PB5 – More reusable code
- PB6 – Increased flexibility in development

Perceived limitations (1 – not at all; 4 – to some extent; 7 – to a great extent)

- PL1 – Inadequate project requirements
- PL2 – Poor project documentation
- PL3 – Inadequate testing
- PL4 – More defects/bugs
- PL5 – Steep learning curve^a
- PL6 – Lack of standards
- PL7 – Instability of requirements
- PL8 – Longer project turnaround time

Perceived hindrances (1 – not a problem; 4 – to some extent; 7 – to a great extent)

- PH1 – Lack of formal guidelines^a
- PH2 – Organizational resistance
- PH3 – Management apathy
- PH4 – Lack of peer support

Subjective norm (1 – not at all; 4 – to some extent; 7 – to a great extent)

SN1 – Your organization^a

SN2 – Peer groups

SN3 – Professional organizations

SN4 – Trade publications

Training (1 – not at all; 2 – 1–4 days; 3 – 1–2 weeks; 4 – 3–4 weeks; 5 – 2–5 months; 6 – 6–12 months; 7 – >1 year)

TR1 – Self-paced course^a

TR2 – Peer mentoring

TR3 – Professional development

Experience

EX1 – Years of software development experience

Organization size

OS1 – Annual revenue (in millions)

OS2 – Number of employees

OS3 – Number of software development professionals

Agile use (1 – not at all; 4 – to some extent; 7 – to a great extent)

AU1 – Extreme programming

AU2 – Agile unified process^a

AU3 – Pair programming

AU4 – Test first

^a Item was dropped because of low loading (<0.707) on its latent construct.

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